CS 57800 – Statistical Machine Learning

Lecture 3 Probabilistic Inference and Statistical Physics

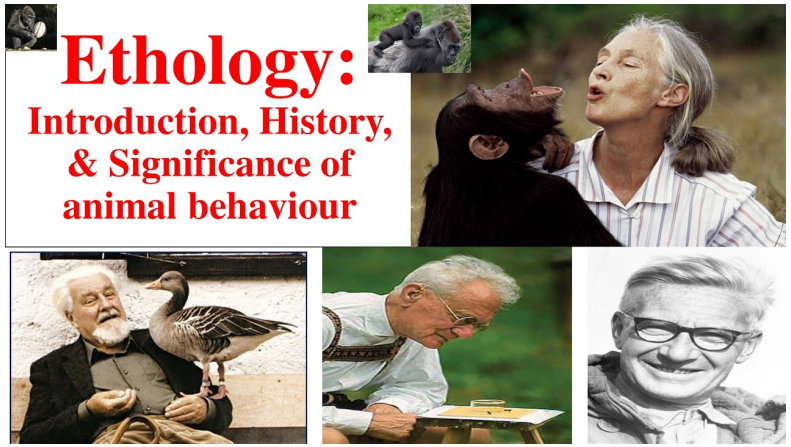
https://ruizhezhang.com/course_fall_2026.html

“Physics” of ML/AI

Math

- Fully rigorous theorems!
- Too idealistic assumptions
- Too simple models (e.g., 2-layer transformer)
- Hard to connect to practice
- Progress very slow

“Ethology”



“Physics” of ML/AI

Math

- Fully rigorous theorems!
- Too idealistic assumptions
- Too simple models (e.g., 1-layer transformer)
- Hard to connect to practice
- Progress very slow

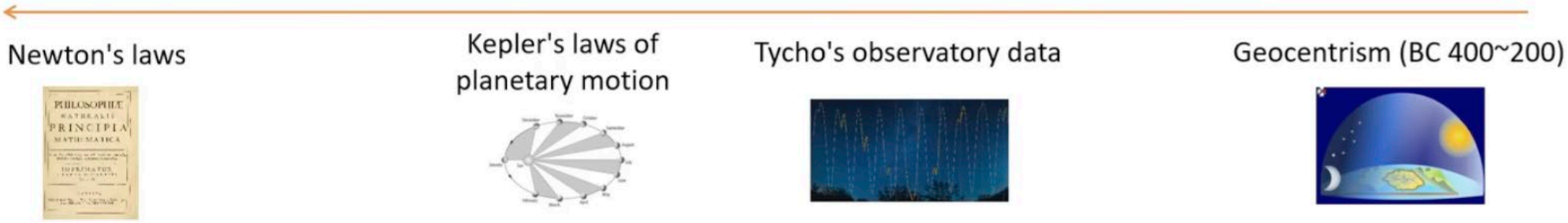
“Ethology”

- Can study large models (GPT 6, Llama 4, …)
- Can be very educational (CoT, ToT, MoE, …)
- Less rigorous, too subjective
- Sometimes lead to false claims

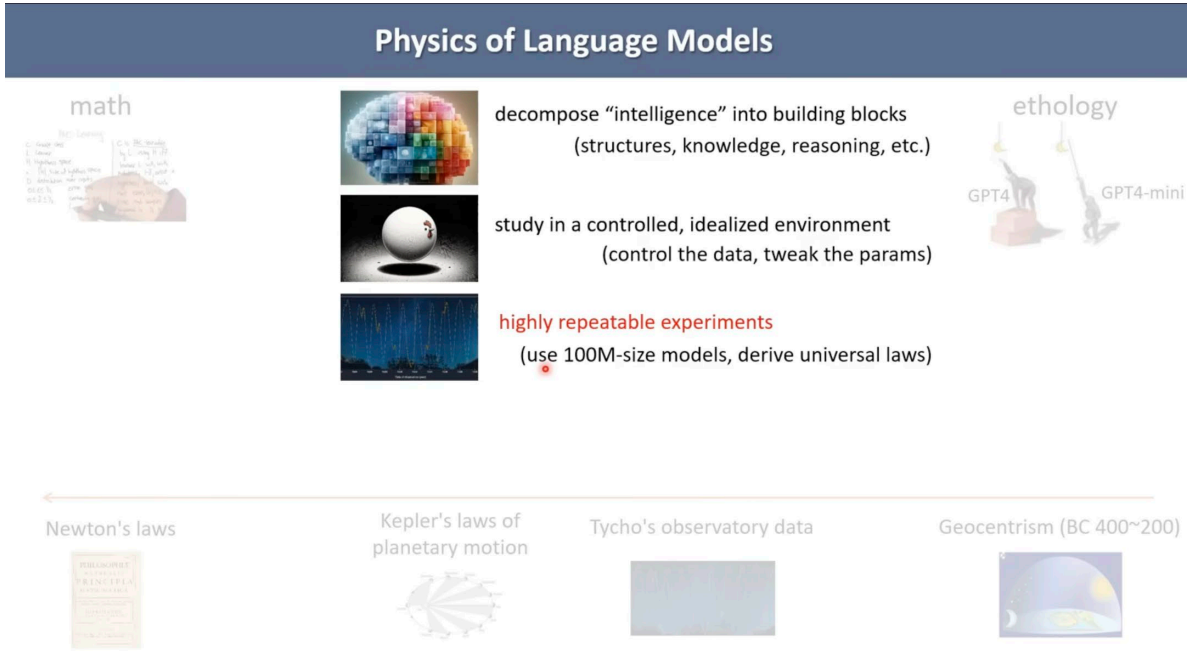
“Physics” of ML/AI

Math

“Ethology”



“Physics” of ML/AI



Physics of Language Models Zeyuan Allen-Zhu

(https://physics.allen-zhu.com/)



ICML 2024
International Conference On Machine Learning

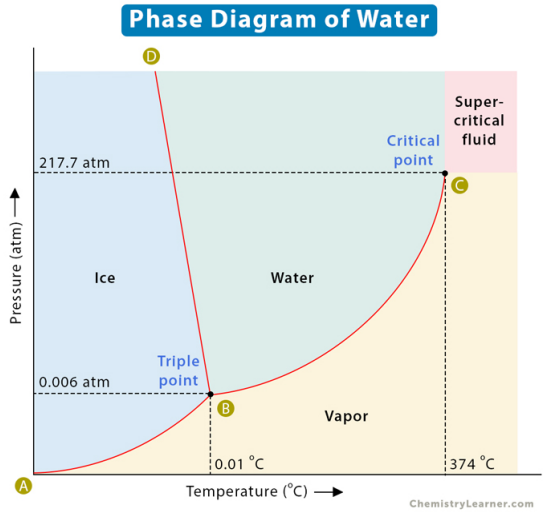
“Physics” of ML/AI

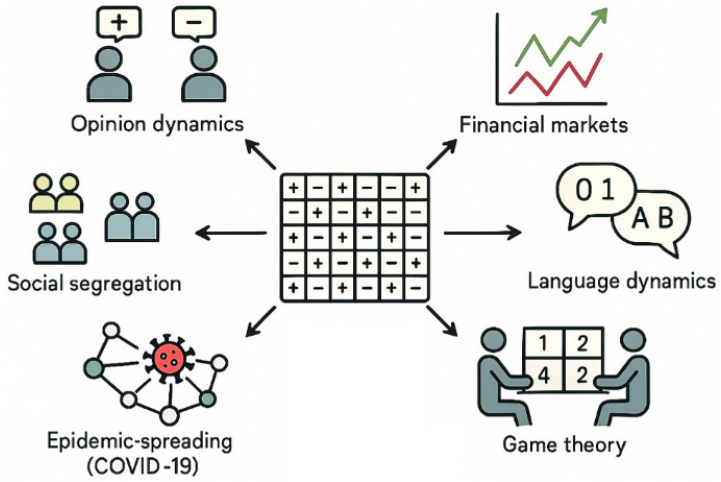
- Our world is super complicated
- Theoretical physicists are very good at working on toy models to understand the world

Example: the Ising model

$$H_N(\boldsymbol{S}) = -J\sum_{\{i,j\}} S_i S_j - h\sum_i S_i\,, \qquad \boldsymbol{S} \in \{-1,1\}^N$$

- Can demonstrate phase transitions (e.g., water freezing into ice, a metal losing its magnetism when heated, etc.)



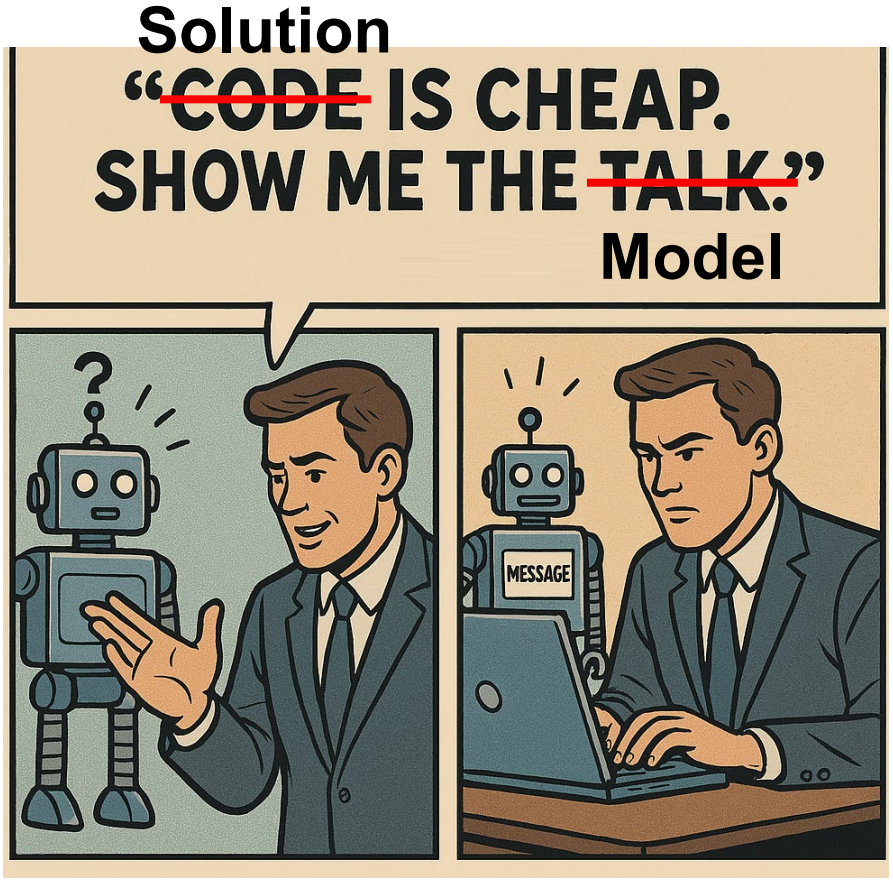


“Physics” of ML/AI

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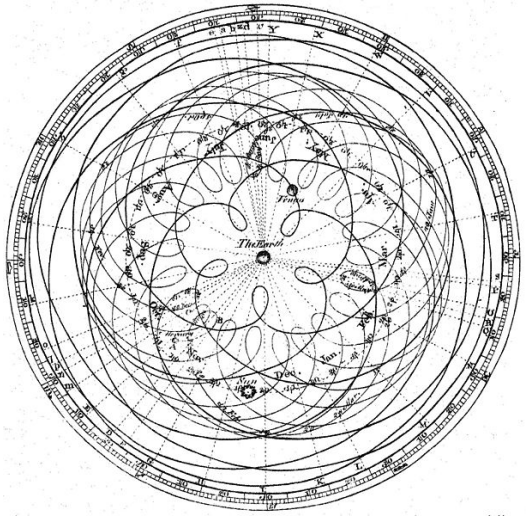
Personal thought:

- In the AGI-era,
- solving a model is much cheaper,
- identifying a useful model (≈problem) is more valuable



ML for Physics

- Physicists have done “machine learning” (inference) over centuries



The Ptolemaic epicycle model

(Overfitting)

Linear regression,
Newton’s method,
etc.

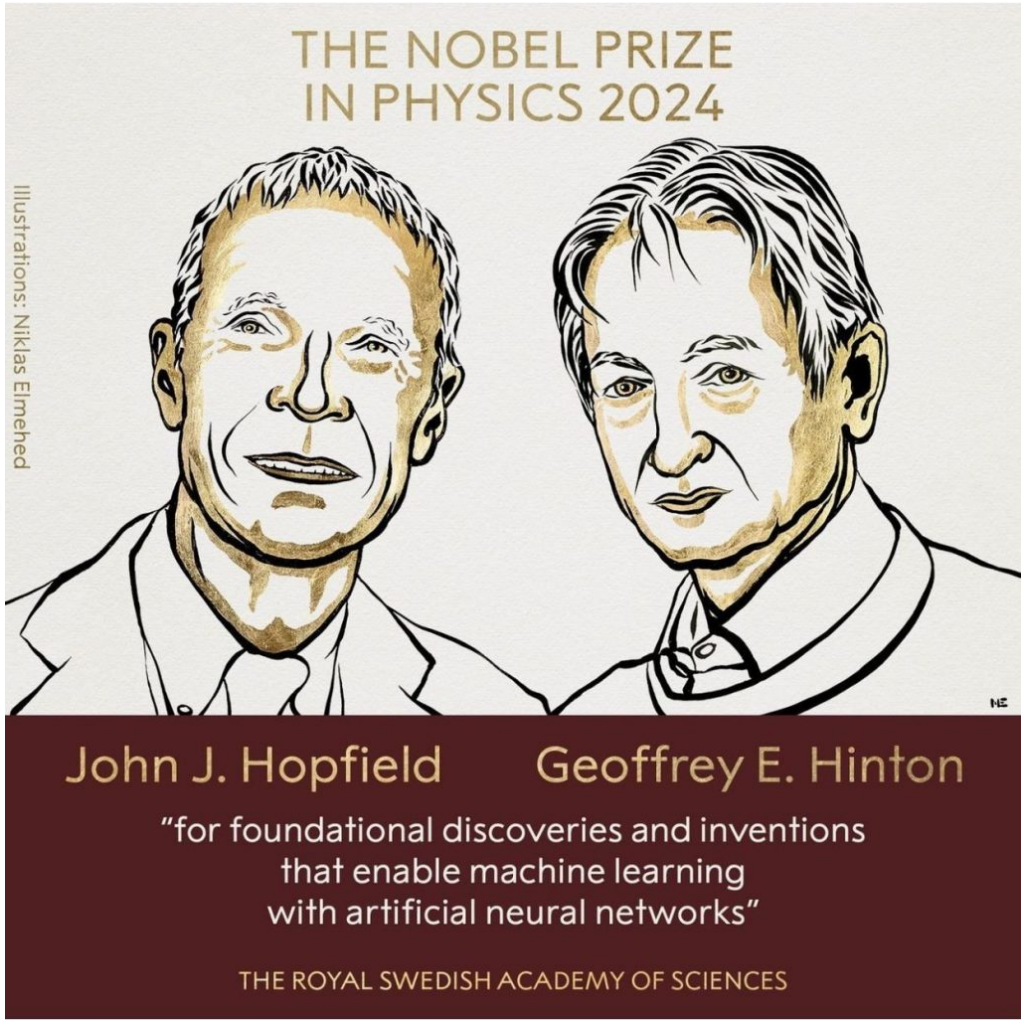
New data

$$F = G\frac{m_1 m_2}{r^2}$$

Newton's law of universal gravitation

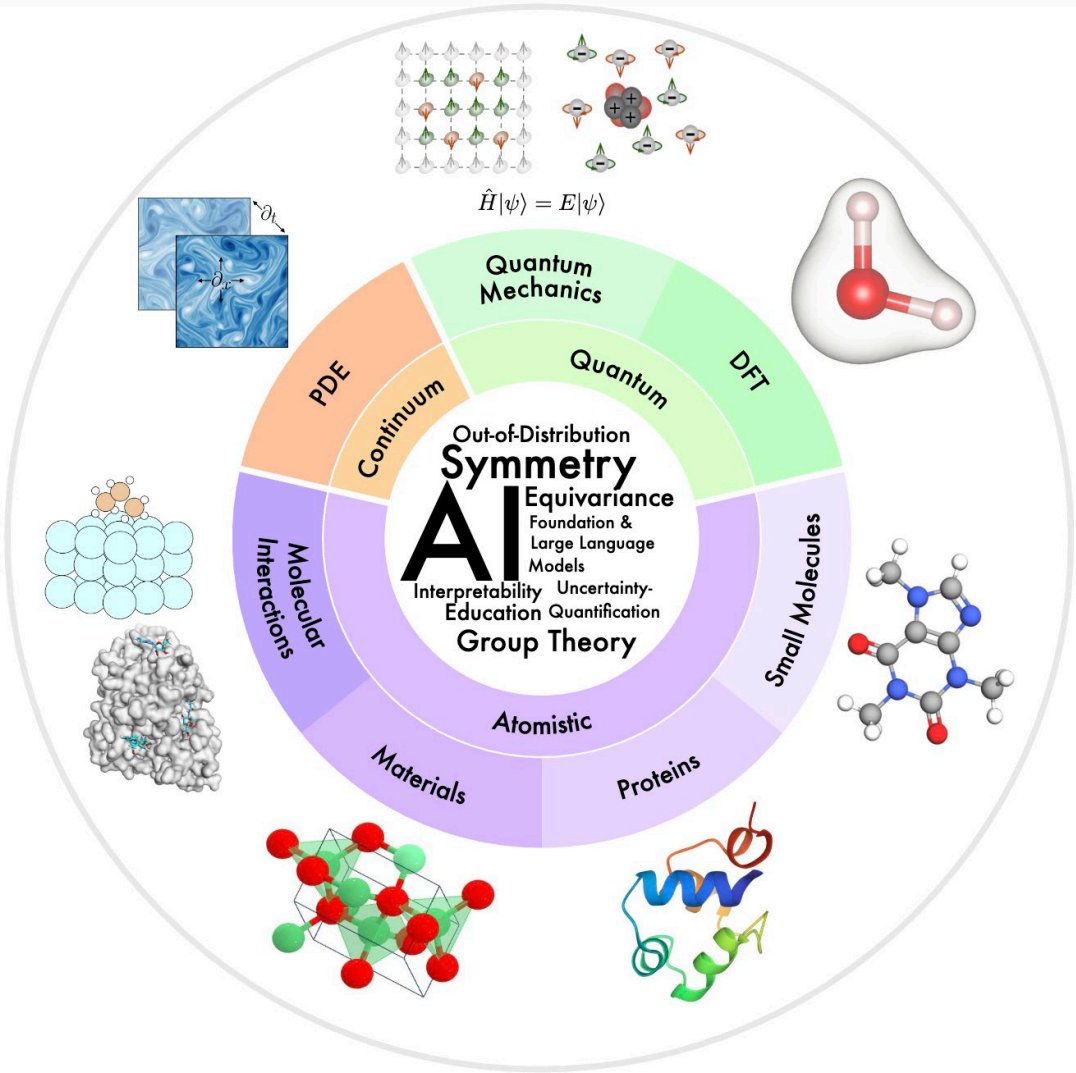
ML for Physics

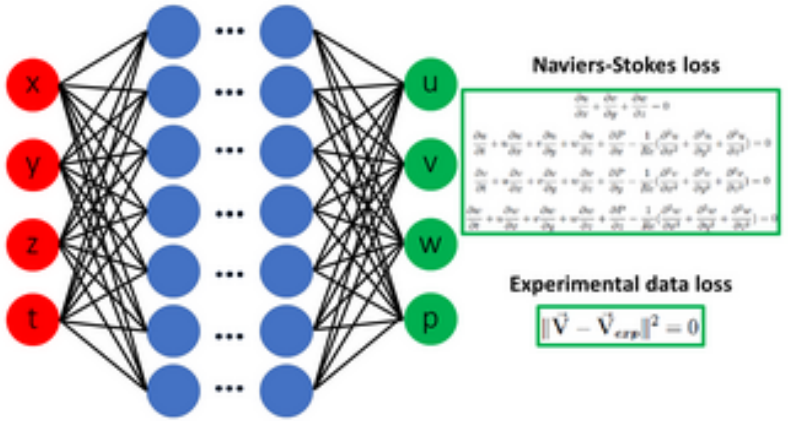
Hopfield network

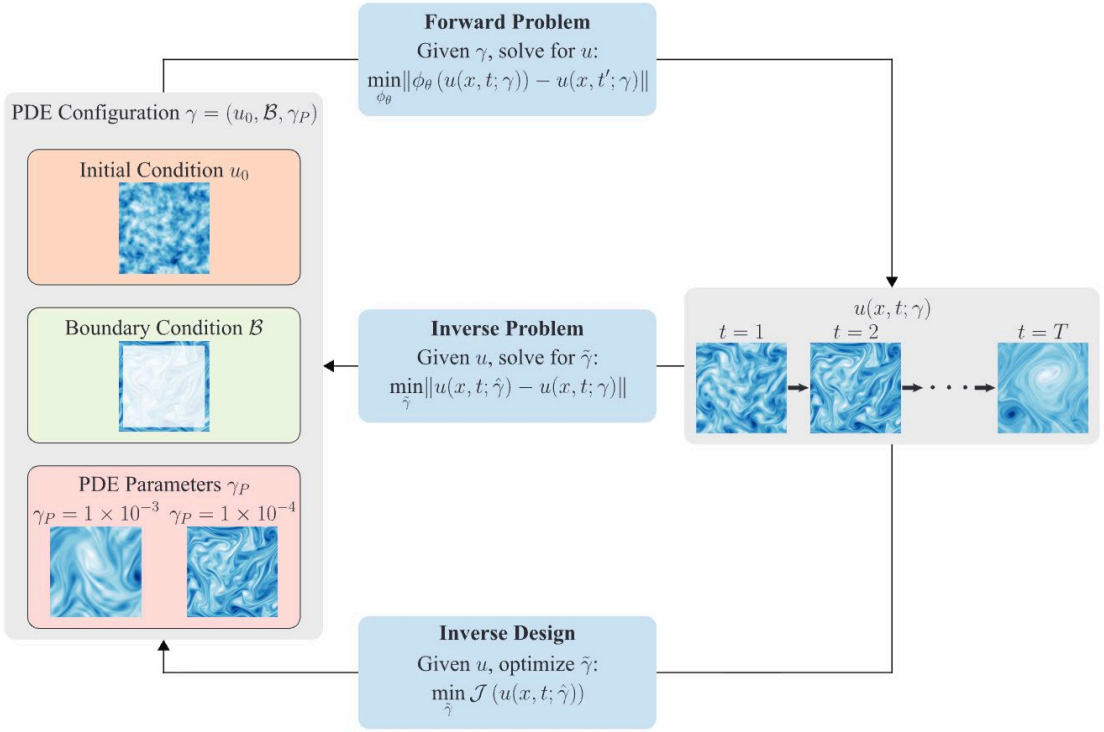


Boltzmann machine

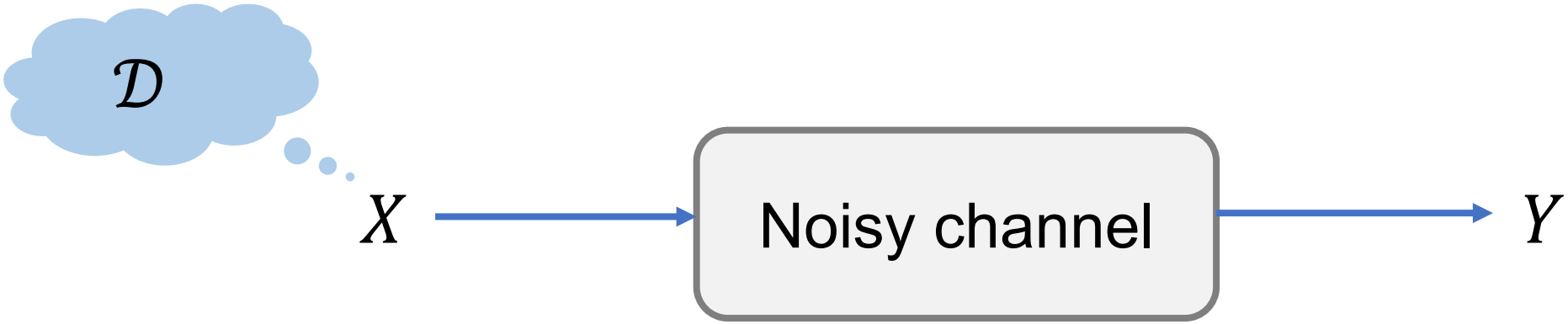
ML for Physics







Physics & Inference



Observing Y, how can you infer X?

- **Teacher-student model for neural network learning:** X is the weight of a neural network F_X. Giving data $\{(x_i, y_i) : x_i \sim \mathcal{N}(0, I_d), y_i = F_X(x_i)\}$, can you learn the neural network weight X?
- **Image denoising:** X is an image from a distribution $\mathcal{D}$, Y is the image with added noise to pixels. Can you recover the original image?

Physics & Inference

The most widely used inference method is the Bayes rule:

$$\Pr[X = x|Y = y] = \frac{\Pr[Y = y|X = x]\Pr[X = x]}{\Pr[Y = y]}$$

- $\Pr[X = x]$: our prior on X
- $\Pr[Y = y|X = x]$: likelihood of the noisy channel
- $\Pr[Y = y]$: normalization constant (or called partition function in physics)

$$Z \coloneqq \Pr[Y = y] = \int \Pr[Y = y|X = x]\Pr[X = x]\,\mathrm{d}x$$

- **Boltzmann energy distribution:**

$$\Pr[X = x|Y = y] = \frac{1}{Z}e^{-\mathcal{E}(x)}$$

$$\mathcal{E}(x) \coloneqq -\log(\Pr[Y = y|X = x]\Pr[X = x])$$

References

- Marc Mezard and Andrea Montanari. *Information, physics, and computation.* 2009.
- Florent Krzakala and Lenka Zdeborová. *Statistical Physics Methods in Optimization and Machine Learning: An Introduction to Replica, Cavity & Message-Passing techniques.* 2022.
- Martin J. Wainwright and Michael Irwin Jordan. *Graphical models, exponential families, and variational inference.* 2008.
- Andrea Montanari and Subhabrata Sen. *A short tutorial on mean-field spin glass techniques for non-physicists.* 2022.

Undirected Graphical Models

- Let $\{\psi_{ij}\}_{(ij)\in F}$ be the compatibility function $\{-1,1\}^2 \to \mathbb{R}_{\geq 0}$ that dictate interactions between particles
- F is a family of edges
- The Gibbs measure μ is a distribution over $\{-1,1\}^n$ given by

$$\mathcal{E}(\boldsymbol{x}) = -\sum_{(ij)\in F} \log\left(\psi_{ij}\left(x_i, x_j\right)\right),$$

$$\mu(\boldsymbol{x}) = \frac{1}{Z} e^{-\mathcal{E}(\boldsymbol{x})} = \frac{1}{Z} \prod_{(ij)\in F} \psi_{ij}\left(x_i, x_j\right),$$

$$Z = \sum_{\boldsymbol{x}\in\{-1,1\}^n} e^{-\mathcal{E}(\boldsymbol{x})}.$$

- $\boldsymbol{x} \in \{-1,1\}^n$ is a configuration of spins

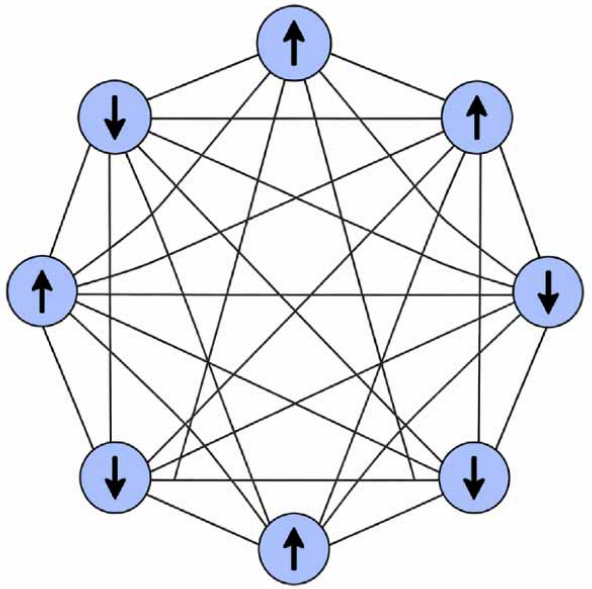
Undirected Graphical Models

Ising models

- $\psi_{ij}(x_i, x_j) = \exp(\beta x_i A_{ij} x_j)$ for some matrix A such that $A_{ii} = 0 \; \forall i \in [n]$ (why?) and A is symmetric

$$\mu(\boldsymbol{x}) \propto \exp\left(-\frac{\beta}{2}\boldsymbol{x}^\top A \boldsymbol{x}\right)$$

- β is the inverse temperature
- A is the Hamiltonian or the interaction matrix of the system
- $\mathcal{E}(\boldsymbol{x}) = -\frac{\beta}{2}\boldsymbol{x}^\top A \boldsymbol{x}$ is the energy of the system
- When $\beta \to 0$ (high-temperature limit), $\mu \to \mathrm{Unif}(\{-1,1\}^n)$
- When $\beta \to \infty$ (low-temperature limit), $\mu \to \mathrm{Unif}(\arg\min \mathcal{E}(\boldsymbol{x}))$

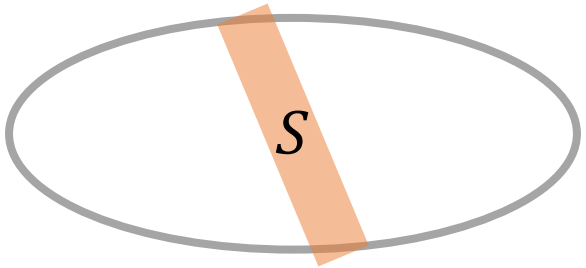


A defines a weighted graph G

Undirected Graphical Models

Markov property

- For $S \subseteq [n]$ such that $[n] \backslash S$ decomposes into disjoint pieces in G, then conditioned on any assignment to the spins in S, the marginal distributions on the pieces are independent



- For Ising models, for any $i \in [n]$, let $S = \partial_i$ (the neighbors of i in G)

$$\Pr\left[x_i = \sigma \middle| \boldsymbol{x}_{\partial_i} = \boldsymbol{s}\right] \propto \exp\left(-\beta\sigma \sum_{j \in \partial_i} A_{ij} s_j\right)$$